

Foundations before performance

Weekly synthesis | 21-27 September 2026 | Q007 | Prepared 27 September

Sidiq review status: Not yet reviewed. Publication is authorised separately.

Finding and scope

The week established reproducible checks for information timing, label overlap, source integrity, data-access declarations and timestamp ingestion, plus the first C++ exercise. Six research items (Q001-Q005 plus Q005a) and C001 are complete. Q006 calendar rules and actual licensed data access remain open. There is no empirical alpha, market-calibrated execution or option-market result.

This synthesis uses all seven daily notes, their validation records, six experiment result files, the C++ fixture and ten commits through 5701e70. It consolidates Sunday's ingestion work without implementing it again. The three September 21 follow-up commits refined role evidence, activated schedules and added the C++ track; they were not three additional research experiments.

What actually landed

Date / commit	Completed work	Decisive evidence
Sep 21 / 34b1bec	Q001 availability	Final-vintage join wrong at 4 of 6 invented decisions
Sep 22 / 981ab66	Q002 label purging	Retains A,D; removes B,C; overlaps fall from 2 to 0
Sep 23 / 85ca7bd	C001 fixed savings	1000 at 5%: interest 50.00, balance 1050.00
Sep 24 / fab2f76	Q003 manifests	Four source hashes match; changed byte rejected
Sep 25 / c4c9b41	Q004 universe/access	Neither candidate admitted; 24 downgrade controls block
Sep 26 / 362d52e	Q005a UTC chronology	Future-fold leak fixed; 16 pre-fix failures resolved
Sep 27 / 5701e70	Q005 strict ingestion	Two valid revisions; all eight bad batches rejected

The other commits are 1ea44bf, b2f1725 and 8e21bdb. Original activity dates and historical notes are retained. Q003 and Q004 happened later than their original targets because Wednesday became a C++ slot and the observed fold defect needed its own bounded correction before ingestion.

The central lesson

A passing component is evidence for its stated contract. It does not validate the next boundary: correct parsing does not authenticate a release, correct hashes do not establish lawful access, and correct accounting does not establish a profitable strategy. The week improved those boundaries by preserving explicit failures and adding independent expected answers.

Concepts to be able to explain

Availability: an observation date describes an economic event; its publication time says when a version could become known. With decision t , publication a , observation o , latency L and maximum age M , require $a \leq t$, $t - a \geq L$, and $t - o \leq M$ when M is configured. Compare UTC instants. Choose the newest observation first, then its newest eligible revision. A revision does not reset age.

In Q001 the correct six-day path is missing, missing, 100, 100, 60, 60. Joining the final value 60 onto every earlier date fails four times. This is a constructed failure count, not a measured market error rate. A fabricated source publication time would still defeat an otherwise correct selector.

Purging: a training row can be old while its target needs a future price. For this past-only holdout starting at a , retain a candidate only when its label end $e < a$. Equality is removed under the closed-label convention. The one-row gap removes safe D while leaving unsafe B,C. Purging removes both unsafe labels, at the cost of reducing training observations from four to two. It does not make overlapping evaluation outcomes independent or replace uncertainty estimation.

Fold and gap: on the London autumn fixture, 01:30 +01:00 is 00:30 UTC and 01:30 +00:00 is 01:30 UTC. Equal clock labels hide 60 elapsed minutes. A fold has two possible instants; a spring gap has none. Q005a fixes comparisons by converting to UTC; Q005 separately checks that the source wall fields and explicit offset return unchanged after conversion through UTC into its declared zone. This distinguishes consistency from truth. The source must still supply the intended interpretation.

The Python datetime documentation and PEP 495 were rechecked for this synthesis. Same-zone comparisons can ignore fold, and constructing a datetime does not itself reject every invalid local label. These are the reasons for separate chronology and ingestion contracts, not universal vendor rules. <https://docs.python.org/3/library/datetime.html> <https://peps.python.org/pep-0495/>

Important negative results

Q004's invented holdings start at 100 each and finish at 120, 90 and confirmed zero recovery. The full return is $210 / 300 - 1 = -30\%$; survivors alone give $210 / 200 - 1 = +5\%$, a 35-point difference. Dropping failures changes the population. Missing terminal prices must never be treated as confirmed zero recovery. These numbers establish arithmetic, not the size or sign of real bias.

The archived CRSP/Norgate assessment admits zero of two candidates: each has eight unresolved project gates. Public product descriptions did not establish entitlement or audited sample quality. The September 25 source review also identified a historical-vintage limitation in current Norgate histories. No source was purchased, downloaded or newly reassessed in this weekly run.

Actual quality evidence and its limits

Evidence	Observed result / scope
Daily final Python totals	25, 49, 49, 94, 144, 167, 257; cumulative, not additive
Weekly make check	257 passed; no added tests or production changes this afternoon
Ruff / formatting / mypy	Passed; 30 formatted files; 16 strictly checked package files
C++ check	Existing executable matches independent expected output
Dependencies	pip --no-cache-dir check: no broken requirements
Archived experiments	Five scripts replayed twice to scratch; every byte matches
Q001 replay	Full-suite artifact test reproduces its archived release schedule
Historical strong controls	100 interval fixtures; 10,800 UTC oracle comparisons; 24 row permutations
Code coverage	Statement and branch percentages not measured; no percentage claimed

The September 21 note reports an initial 21-case core run; its final validation records 25 cases including artifacts. The unchanged 49 on Wednesday excludes the separate C++ executable check. The 10,800 comparisons are cases inside an oracle test, not 10,800 additional pytest tests. Local validation uses the existing Python 3.12 environment on macOS. This run reused the built C++ executable; the daily records separately document warning-clean compilation and a wrong-rate mutation.

At the weekly baseline, authenticated GitHub reads returned zero open issues and a successful workflow for 5701e70. The workflow configures Python 3.11/3.12/3.13. This observation is tied to that commit, not a prediction about the new synthesis commit. Publication status is recorded separately. <https://github.com/SidiqB/quant-research-apprenticeship/actions/runs/36303053768>

Failures that changed the work

The important correctness failure was observed on September 22: a later-fold publication entered an earlier-fold decision. A UTC workaround was documented while C001, manifests and access work continued. September 26 reproduced 16 failing regressions before correcting availability. September 27 then rejected inconsistent offsets and missing local times, which UTC conversion alone did not settle.

Daily checks also caught long source lines and a type-narrowing issue. They were corrected before the reported successful runs. Earlier PDFs spilled tables or answers onto almost empty pages; notes were shortened or given deliberate breaks and inspected again. These are recorded repairs, not evidence that the first attempts passed. One C++ mutation used 5.0 instead of 0.05 and correctly failed the output comparison, demonstrating a percentage-unit error rather than general correctness.

This weekly run initially hit restricted-network DNS failure; an approved fetch succeeded and main was already current. A shell inspection accidentally reused zsh's path variable, causing command lookup failures in that subprocess; repeating it with a task-specific variable recovered. No project files were affected. pip disabled an unwritable cache; the explicit no-cache dependency check passed. The GitHub CLI is absent, so the read-only API check used existing authentication without logging it.

Employer-skill connections

The stored September 21 cohort contains 49 deduplicated postings across eight firms. It is purposive, includes junior and senior jobs and is not a census or an eligibility assessment. Python appears in 37 postings, statistics and communication in 29 each, and C++ in 13. These are explicit mentions, not counts of mandatory requirements. No vacancy availability was refreshed this week.

Research infrastructure: the stored AQR data-scientist evidence names validation, Git, PyTest and CI/CD; Jane Street's research-engineer evidence emphasises reproducibility and research infrastructure. Manifests, negative controls and source-zone validation supply relevant repository examples. They do not establish production-scale experience. Sources from the dated cohort:
https://careers.aqr.com/jobs?gh_jid=8088184
<https://www.janestreet.com/join-jane-street/apply/7437779002/>

Research judgement and communication: the changing-universe counterexample and interval proof show how to explain a failed assumption with a hand-worked example. This is useful preparation for systematic research and model-validation discussions. There is still no fitted model, uncertainty estimate, out-of-sample alpha or independent personal reproduction to present as completed evidence. The stored Citi model-validation role is a future benchmark for accuracy and independent comparison:
<https://jobs.citi.com/job/new-york/model-validation-2nd-lod-sr-lead-analyst/287/100430304704>

C++ development: C001 connects compilation, numeric units and a known-answer check. C002 should add a reusable function with a declared compounding convention. One fixed example does not demonstrate low-latency systems, ownership design or performance engineering. The dated IMC role is a skill benchmark: <https://job-boards.eu.greenhouse.io/imc/jobs/4945420101>

The role-analysis wording is corrected to distinguish Sidiq's authorised beginner C++ learning track from replacing research components for speed. Only the latter still needs a measured bottleneck, a validated reference and evidence of benefit. The employer sample and its historical counts are unchanged.

What remains unproven

No admitted market dataset, historical receipt audit or verified current-source entitlement exists. The adapter returns UTC observations without retaining raw source text or zone provenance and checks duplicates within a batch. Calendar membership, persistent cross-batch uniqueness and vendor-file integration remain separate work. Named-zone tests rely on installed timezone rules; the dependency lock does not capture the whole operating system or IANA database. A manifest's four hashes identify selected files, not a complete environment or authenticated authorship.

No personal review or independent understanding is inferred from automatic test success. Sidiq should be able to derive one counterexample and reproduce it before describing it as personal interview work.

Revised priorities: 28 September-4 October

Slot	Task / dependency	Acceptance focus
Mon Sep 28	Q006a, after Q005	Calendar source, bounded session fixture, 09:00 decision policy
Tue Sep 29	Q006b, after Q006a	Session lookup, holidays, shortened sessions, boundary tests
Wed Sep 30	C002, after C001	Reusable compounding function, zero cases, invalid inputs
Thu Oct 1	Q008, after Q006b	Hand-check split and dividend total-return arithmetic
Fri Oct 2	Q009a, after Q008	Historical membership intervals; no current-list filtering
Sat Oct 3	Q009b, after Q009a	Known terminal proceeds versus unresolved exit evidence
Sun Oct 4 morning	Q010, after Q009b	Lagged returns invariant to added future observations
Sun Oct 4 at 16:00	Q014 synthesis	Reassess data readiness and conditional phase dates

These are targets, not promised completion dates. Each is one 45-90 minute daily step; unfinished work retains precedence and Wednesday keeps its C++ slot. Q011 and Q012 move to October 5 and 6; Q013 moves to October 8 around C003 on October 7. Q015 onward retains original phase markers only, with actual dispatch following dependencies. Phase 1's first week now builds synthetic mechanics; its empirical portion remains blocked. Later phase windows are conditional, not silently compressed.

Backlog changes and changed assumptions

Q006 is split into a documented calendar/decision contract (Q006a) and implementation (Q006b). A timestamp can be valid before the exchange opens: the provisional 09:00 New York decision must be defined separately from regular-session execution. Do not reject it merely for being pre-open or imply that a decision-time close is an executable fill. Holidays and early closes need source evidence.

Q009 is split into membership intervals (Q009a) and terminal-event treatment (Q009b). The -30%/+5% fixture shows why population membership and exit proceeds must not be collapsed into one shortcut. Q004a becomes an explicit parent of Q004a1 entitlement/output permissions and Q004a2 licensed sample audit. Both remain blocked in sequence; no declared gate becomes verified just because it has a subtask.

Q005b adds provenance retention and a two-batch integration check after Q013, before empirical use. Q005 remains complete within its in-memory scope. Preserve source hashes, zone, rule version and duplicate behavior at integration; this split does not authorise a vendor acquisition.

Input needed from Sidiq

Before acquisition, identify existing licensed access or a source/budget preference; do not send credentials. If historical vintages are unavailable, should a narrower retrospective study be considered? That change needs explicit agreement. Also identify the separate finance project mentioned September 21 when convenient. Neither answer blocks the next synthetic step or authorised pushes.

Revision exercises

1. A value belongs to January 1 but is published January 3 and revised January 5. Explain the value available on January 2, January 4 and January 6, and why the final-vintage join is invalid.
2. Evaluation starts January 5. Why must a January 2 training label ending exactly January 5 be removed here? Why did removing the latest training row fail in the eight-row fixture?
3. Map London's two October 25 01:30 labels to UTC. At the later instant, does a first-fold release meet a 60-minute latency, and can a new revision rescue a 59-minute maximum observation age?
4. What do a matching hash, a valid source-zone round trip and a passing access declaration each prove? Identify one thing each leaves unproven.
5. Why can the 1000-at-5%-for-one-year C++ fixture not distinguish simple interest from compounding? State the two-year outcomes and the convention C002 must declare.

Suggested answers

1. Missing, 100 and 60 respectively. An economic date is not an information-arrival date. The revised value cannot be supplied to decisions before publication. Adding genuinely future data must not alter earlier valid snapshots.
2. Under the declared closed-label boundary, the target's last input is not established before the first evaluation decision. Retain only $e < a$. Row D is recent but safe; B and C are older decisions with later endpoints. A fixed row count therefore removed the wrong information in this fixture.
3. The offsets +01:00 and +00:00 map to 00:30 and 01:30 UTC. Exactly 60 elapsed minutes satisfies the inclusive latency boundary. The observation is still 60 minutes old and fails the 59-minute age limit even after revision; publication time does not reset economic age.
4. The hash matches selected bytes against a stored digest, not authorship or truth. A round trip checks clock-rule consistency, not actual publication. A declaration checker validates required statuses and references, not whether the supplied entitlement or sample evidence is authentic.
5. Both formulas equal $P * (1 + r)$ at one year. At two years, simple interest gives 1100 and annual compounding gives 1102.50. C002 must declare the period/rate and horizon contract and validate it; formatting to two decimals does not provide exact decimal-money arithmetic.

Evidence trail and reproduction

Read `research_log/2026-09-21.md` through `2026-09-27.md` alongside `reports/milestone/initial-validation.md` and the six dated validation records. Results are `experiments/availability`, `purging`, `manifests`, `data_access`, `timezones` and `ingestion/results.json` (each directory contains its own `results.json`). C++ evidence is `cpp/projects/savings_growth/expected.txt`. Role sources and caveats are in `career_research/role_skill_analysis.md` and `career_research/research_methodology.md`.

Run `make check` and `.venv/bin/python -m pip --no-cache-dir check`. The five newer experiment scripts accept `--output PATH` for scratch reproduction; Q001's artifact test verifies its default result. Render this report with `.venv/bin/python scripts/render_note.py reports/weekly/2026-09-27-synthesis.md reports/weekly/2026-09-27-synthesis.pdf`. Weekly artifact checks are recorded in `reports/weekly/2026-09-27-validation.md`.

Programme scope still ends on 31 December 2026. After that date stop and request the next scope; do not infer permission for a 2027 programme. The existing Europe/London schedule is unchanged.